MIL-C-48171 (AR)
AMENDMENT 3
4 November 1981
SUPERSEDING
AMENDMENT 2
4 October 1976

MILITARY SPECIFICATION

CARTRIDGE, 81MM, HE, M374A3
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-48171 (PA), dated 14 January 1975, and is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, to the list of Military Specifications, add the following:

"MIL-C-48156 - Cartridge, Ignition, M299, Loading, Assembling and Packing.

Following line 21: Add "9312648 - Cartridge, 81MM, HE, M374A3 W/O Fuze".

Line 22: Drawing "9241291" delete "W/Fuze, PD, M524A6" and substitute "W/Fuze, PD, M567".

Line 24: Delete "9230175 - Container...in Jungle Wrap" and substitute "9276596 - Container, Ammunition, Fiber, M252A5 for 81MM Mortar Ammunition".

At bottom of page: Add "9287603 - Packing and Marking for Container, Fiber, M252A5, Ammo for 81MM Ammo in Jungle Wrap".

PAGE 2

3.2, line 2: Add "or 9312648, whichever is applicable".

PAGE 3

* 3.5.2 Delete "function" and substitute "function high order".

FSC 1315

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PAGE 5

4.2.2, line 4: Delete "Inspection for Critical Defects and Major Defects, when so specified)" and substitute "Inspection for Critical Defects and Special Defects (and Major Defects, when so specified)".

Line 12: Examination, insert the following before the last sentence of the paragraph:

"The contractor may elect to perform 100% inspection with automated equipment or impose mandatory process controls on automated operations in lieu of conducting sampling inspection for defects listed in this specification. In such cases, calibration plans shall be submitted to the Technical Agency (see 6.8) for approval for each automatic inspection station and process control plans must be submitted for approval for each of the automated operations. The plans shall identify; (1) the frequency (production interval) for validating equipment functioning, (2) the defect standards and (3) a description of material control procedures to be imposed on product passing through each automated station during the production interval between validations. Under these provisions final acceptance will be withheld from material produced during a production interval until satisfactory validation of each of the automated stations so controlled. In the event of a validation failure of any station, the material processed through such station since the last acceptable validation shall be placed in a rejected status pending corrective action."

PAGE 6

4.2.2.1, line 2: delete "9241291)". at end of line and Add "9241291 or 9312648".

4.2.2.2, line 3: Delete "9241291" and Add "9241291 or 9312648".

4.2.2.2, line 8: Delete "101. Depth...max...Gage CD" add new category heading, "Major: 100 percent inspection 101. Depth of fuze well, max...Gage CD".

4.2.2.3, line 2: Delete "9241291" and Add "9241291 or 9312648".

Line 6: Delete "Critical 2. High explosive extends beyond top of liner...Visual" in its entirety.

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4.2.2.4, line 1: Delete in its entirety and add "4.2.2.4 Cartridge (see dwg. 9241291 or 9312648)".

Line 3: Delete "Critical 1. Ignition cartridge head not fully seated...Visual" and add "Critical 1. Depth of ignition cartridge from fin end below minimum...Gage CD".

PAGE 7

4.2.2.4, line 1 from top of page; Delete line in its entirety and add "2. Fuze safety wire missing or insecure (if applicable; see dwg. 9241291)... Visual-Manual".

Line 5 from top of page: Delete line in its entirety. And add "5...Fin blade distorted...gage".
"6...Cracks in fin housing...Visual (4 power magnification)".

Following line 5 from top of page: Add "Special:
a. Warning label(s) missing...Visual (see 6.7).

Line 10: Delete line "103. Outside diameter, max... Gage CD".

Line 20 from top of page: Delete line in its entirety and add "108. Fuze fails to meet mimimum assembly torque (if applicable; see dwg. 9241291)...Test".

Line 24 from top of page; Delete line in its entirety and add "111. Fuze not fully seated (if applicable; see dwg. 9241291)...Visual.

Line 28 from top of page: Delete "Minor: 202. Pettman cement does not cover...Visual" in its entirety and substitute "Minor: 202. Pettman cement (or alternate) improperly applied (applicable to the fuze, fin and ignition cartridge)...Visual".

Line 33 from top of page add "Minor 205 Closing plug not fully seated (if applicable; see dwg. 9312648) ...Visual.

4.4.2.5, line 1: Delete "see dwg. 9230175" and substitute "see dwg. 9276596".

Line 3: Delete "Critical: None defined" and add "Critical: 1. Propelling charge supports missing... Visual".

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4.2.2.6, line 1: Delete "see dwg. 9230175" and substitute "see dwg. 9276596".

4.2.2.7, line 1: Delete "see dwg. 9230175" and substitute "see dwg. 9287603".

Following line 10: Add 203. Kraft paper missing or not properly assembled...Visual".

Add the following paragraph:

"4.2.2.9 Liner (See dwg. 7549011).

Categories	Defects	Method of Inspection
Major 101.	100 percent inspection Length, min.	Gage
Major 102.	Length, max.	Gage
103.	Outside diameter, min.	Gage

Add the following paragraph:

"4.2.2.10 Packing and Marking for Box, Wirebound Packing, Ammunition for 81MM Mortar Ammunition in Jungle Wrap (see dwg. 9287601).

Categories	Defects	Method of Inspection
Minor 201.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3.1, line 2: Delete "XM299" and substitute "M299".

4.2.3.2: Delete entire paragraph and substitute "Cavitation (see 3.3 - except 3.3.4) Major Defect - The sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-1235 using an AQL of 0.65 percent. Any cartridge having cavitation in excess of that permitted by the applicable requirement for any segment shall be classed defective. The test shall be performed as specified in 4.3.1."

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PAGE 10

4.2.3.4.2, line 2: Delete "4.2.3.3.1" and add "4.2.3.4.1".

PAGE 11

4.3.2, line 5: Delete "One half of the sample cartridges shall be fired at charge one and the other half at charge nine." and add "One half of the sample cartridges shall be fired at charge zero and the other half at charge four".

Line 8: Delete "The individual velocities, pressures and ranges shall be recorded on the firing record" and substitute "The individual velocities, pressures (charge four only), deflections and ranges shall be recorded on the firing record".

Line 10: Delete "In the event of a misfire, the ignition cartridge shall be removed and forwarded to Picatinny Arsenal" and substitute "In the event of a misfire, the ignition cartridge shall be removed and forwarded to Picatinny Arsenal, ATTN: DRDAR-LCE-E".

Line 31: Delete "...and prints of all the other rounds in that particular charge zone, shall be attached to the firing record." and add "...and prints of all the other rounds in that particular charge zone, in that lot, shall be attached to the firing record."

PAGE 12

5.2: Delete this paragraph in its entirety and substitute "5.2 Packaging, packing and marking - Packaging, packing and marking shall be in accordance with dwg. 9287603, 9287601, 9276596".

PAGE 13

6.2.2, line 15: Delete "The Contractor if referred to MIL-HDBK-204, 'Inspection Equipment Design' for guidance."

Line 17: Delete "five digit" and substitute "paragraph and".

6.3, line 6: Delete this line in its entirety and add "ATTN: DRDAR-QAR-Q, DRDAR-LCU-D."

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Add the following paragraph:

6.7 Special Defect.-A defect, other than Critical, that judgement and experience indicate may, depending upon the degree of variance from the design requirement:

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. Prevent performance of the tactical function of a major end item.

Add the following paragraph:

6.8 Submission of Calibration and Process Control Plans.- Prior to implementation of calibration or process control plans in accordance with 4.2.2, plans shall be approved by Picatinny Arsenal, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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